IS1212 : Probability and Statistics

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Descriptive statistics for summarizing data

- Use to summarize numerical data.
- Raw data provide little information.
- Need to convert the raw data into useful information.
  - Age of the patient

Row Data	56	25	45	55	51	24	28	23	37	51	55	62	36	25	46	55	47	40	28
Ordered Data	23	24	25	25	28	28	36	37	40	45	46	47	51	51	55	55	55	56	63

  - Smallest value = 23, Largest value = 63, Range = 40, Most frequent value = 55 Middle value = 45
- Easier to get a clear idea about the data set when it is clear and specific.
- Summarizing data makes easy to understand.
- Summary measurements reduce a data set to an understandable form
- Summary measurements simplify the data which describe the important features.
- Measures of Locations, Measures of Dispersions and Measures of Relationships help to get the attention of your audience.

Summary of Measures

- The **central tendency** is the extent to which all the data values group around a typical or central value.
- The **variation** is the amount of dispersion or scattering of values
- The **shape** is the pattern of the distribution of values from the lowest value to the highest value.

Measures Of Locations

Measures of Locations

- Minimum
  - The first observation of the ordered data set
- Maximum
  - The last observation of the ordered data set
- Measure of Center
  - A quantity locates a center of the data set.
  - The most common measures of center are,

    1. Mean

    2. Median

    3. Mode
- Quartiles
  - A quantities that divide the data set into four equal parts.
- Percentiles
  - A quantities that divide the data set into hundred equal parts.

Mean

- The average value of the observations
- Total divided by the number of values
- Use μ notation for the population mean and "bar" notation for the sample mean

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- Very sensitive with extreme values.
- For a symmetric distribution mean in the middle
- In a skewed distribution without outliers mean move towards to long tail.
- There are different version of the mean
  - Arithmetic mean (ordinary mean)
  - Weighted Mean
    - When we wish to give greater emphasis on some values
    - The weighted mean with weights $w_1, w_2, w_3, \ldots, w_n$ is: $\bar{X}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$

The Median (M)

- The value which divide the ordered data set into two equal parts
- Half of the observations are smaller and other half are larger than the M.
- Steps to find the median
  1. Arrange all observations in order (the smallest to the largest)
  2. If the number of observation (n) is;
     - odd, then the center value:

$$M = \frac{(n+1)^{th}}{2} \text{ value}$$

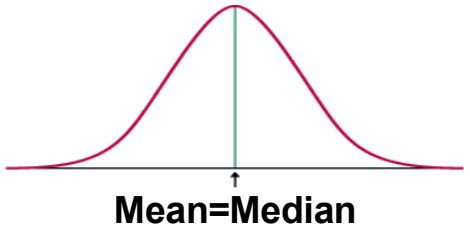
     - even, then mean of the two middle values in the ordered list:

$$M = \frac{\left(\frac{n}{2}\right)^{th} value + \left(\frac{n+1}{2}\right)^{th} value}{2}$$

Mean Versus Median

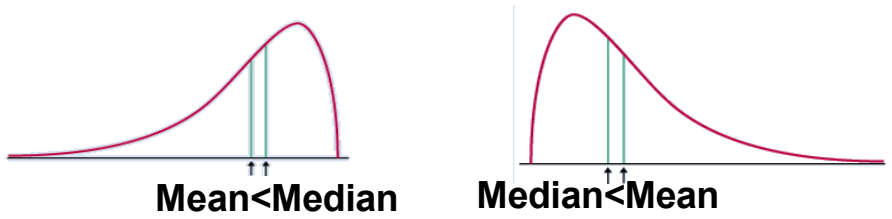
- Mean is the informal midpoint of the data set.
  - May or may not the data set divide into two equal parts.
- Median is the formal midpoint of the data set.
  - Exactly at the middle, data set divided into two equal parts.
- The mean and the median are the same only if the distribution is symmetrical.
- The median is a measure of center that is resistant to skew and outliers. The mean is not.
- In a skewed distributions the mean is further out in the long tail than the median.

Mean and median for a symmetric distribution



Symmetric

Mean and median for skewed distributions

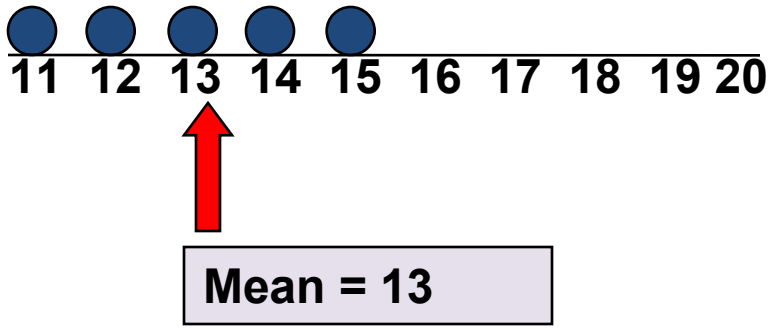


Left skewed

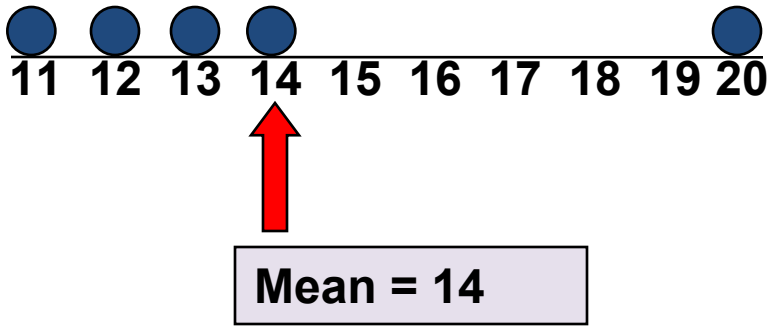
Right skewed

Mean Versus Median

The most common Mean is **Affected** by extreme values (outliers).



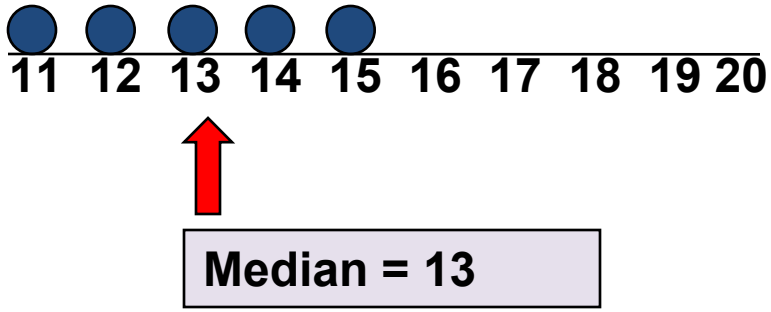
$$\frac{11+12+13+14+15}{5}=\frac{65}{5}=13$$

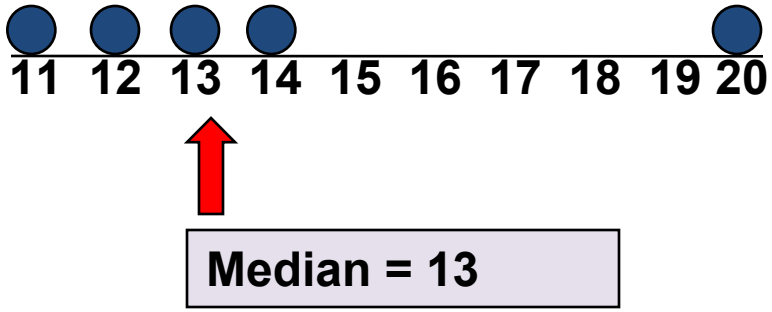


$$\frac{11+12+13+14+20}{5}=\frac{70}{5}=14$$

Mean Versus Median

The median is the "middle" number (50% above, 50% below) which is **Not affected** by extreme values.





The Mode

- Most frequent observation.
- There may be more than one mode

Value	20	**30**	40	**50**	**60**
Count	5	**10**	2	**10**	**10**

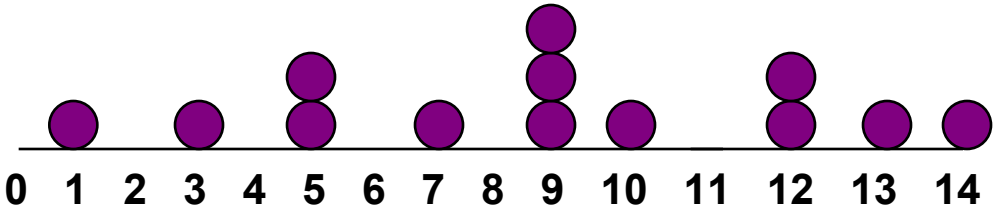
- There may be a data set without a mode

Value	20	30	40	50	60
Count	5	5	5	5	5

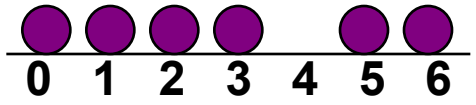
- Less frequently used than the mean or the median
- The only measure used for both categorical and numerical data

Blood Group	O+	O-	B+	B-	A+	A-
Count	5	10	12	2	11	8

The Mode



Mode = ?



Mode = ?

Resistant Measure

- Does not influence by extreme values.
- Does not respond strongly to changes in a few observations, no matter how large these changes.
- Mean is not a resistant measure
- Median and mode are resistant measures

Quartiles

- Divide the ordered data into four equal subsets.
  - First quartile; $Q_1 = 1(n+1)/4^{th}$ value
  - Second quartile; $Q_2 = 2(n+1)/4^{th}$ value
  - Third quartile; $Q_3 = 3(n+1)/4^{th}$ value

Percentiles

- Divide the ranked data into 100 equal subsets.
  - The k-th percentile (P_k):
    - k% of the data are smaller and (100-k)% of the data are larger.

Notes:

1. $Q_1 = P_{25}$,
2. $Q_3 = P_{75}$
3. Median = $P_{50} = Q_2$

Percentiles…

- A simple procedure to obtain P_k

  Step 1 : Rank n data values, from the smallest to the largest

  Step 2 : Calculate $nk/100$

nk/100

An integer results (a)

Step 3: $d(P_k) = a.5$

Step 4: P_k : Average of ath and (a+1)th values

A number with a fraction results

$d(P_k) = b$, the next larger integer

P_k is the value of the data in the bth position

Example

- Exam marks of 50 students are listed below. Find:
  - The first quartile,
  - The 58th percentile
  - The third quartile.

60	47	82	95	88	72	67	66	68	98
90	77	86	58	64	95	74	72	88	74
77	39	90	63	68	97	70	64	70	70
58	78	89	44	55	85	82	83	72	77
72	86	50	94	92	80	91	75	76	78

Example…

Find Q_1 first.

Step 1 : Rank the data

39 44 47 50 55 58 58 60 63 64 64 66 67 68 68 70 70 70 72 72 72 72 74 74 75 76
77 77 77 78 78 80 82 82 83 85 86 86 88 88 89 90 90 91 92 94 95 95 97 98

Step 2: Find nk/100 = 50*25/100 = 12.5

Step 3: $d(Q_1) = 13$

Step 4 : Q_1 is the 13th value from the lowest value. $Q_1 = 67$

Verify that $P_{58} = 77.5$ and $Q_3 = 86$.

Measures Of Dispersion

Measure of Spread (Dispersion/ Variability)

- Measures how far each value is away from measure of center
- Without measures of dispersed, measure of center may be misleading.
- For example:

		median	spread of the data
2, 5, 6, 8, 9	:	6	9-2=7
5, 5, 6, 6, 6	:	6	6-5=1
6, 6, 6, 6, 6	:	6	6-6=0

- Measure the **variability** should be
  - a real number
  - zero if all the data are identical
  - increases as the data becomes more diverse
  - cannot be less than zero

Range (R) and Inter Quartile Range (IQR)

- Range : **R = largest value – smallest value**
  - The simplest measure of dispersion
  - Gives an indication of the absolute spread of the distribution
  - Depends only on the two extreme values
  - Not very informative & can be affected by outliers

- Inter Quartile Range (IQR): **IQR = Q_3 - Q_1**
  - **Range of the middle 50%** of the values
  - Upper 25% and lower 25% observations are discarded
  - Not affected by outliers
  - Not centered on the middle unless the distribution is symmetric
  - Often used with median

Deviation

- Sample mean is located in a data set such that the sum of the differences of the numbers from the mean is zero.

$$\sum_{i=1}^{n}\left(x_i - \bar{x}\right) = \sum_{i=1}^{n}\left(differences\right) = 0$$

- The sample mean represents the center of gravity.
- The difference of a given number from the mean is the deviation from the mean.
- The sum of the deviation from the mean is not useful
- We might consider;
  - The sum or mean of the Absolute Deviations
  - Square the Deviations

Variance

- Measure the dispersion by squaring the deviations before averaging them.
- Numerically, the variance equals the average of the several squared deviations from the mean.
- Variance for the sample of size n is given by

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)} = \frac{\sum x^2 - n\bar{x}^2}{(n-1)}$$

- A data set which is very different, will have a larger variance compared to a data set which is relatively homogeneous.

Standard Deviation

- **Positive value of the square root** of the variance.
- If each number is increased by a constant c
  - The mean is increased by c
  - The standard deviation remains unaltered
- If each number is multiplied by a constant k
  - The mean is multiplied by k
  - The standard deviation is multiplied by k

Coefficient of Variation (Relative Variability)

- The coefficient of variation (CV) equals the standard deviation divided by the mean.

$$CV = \frac{s}{\bar{x}}$$

- It can be expressed either as a fraction or a percent.
- Used to compare the scatter of variables expressed in different units.

Coefficient of Variation…

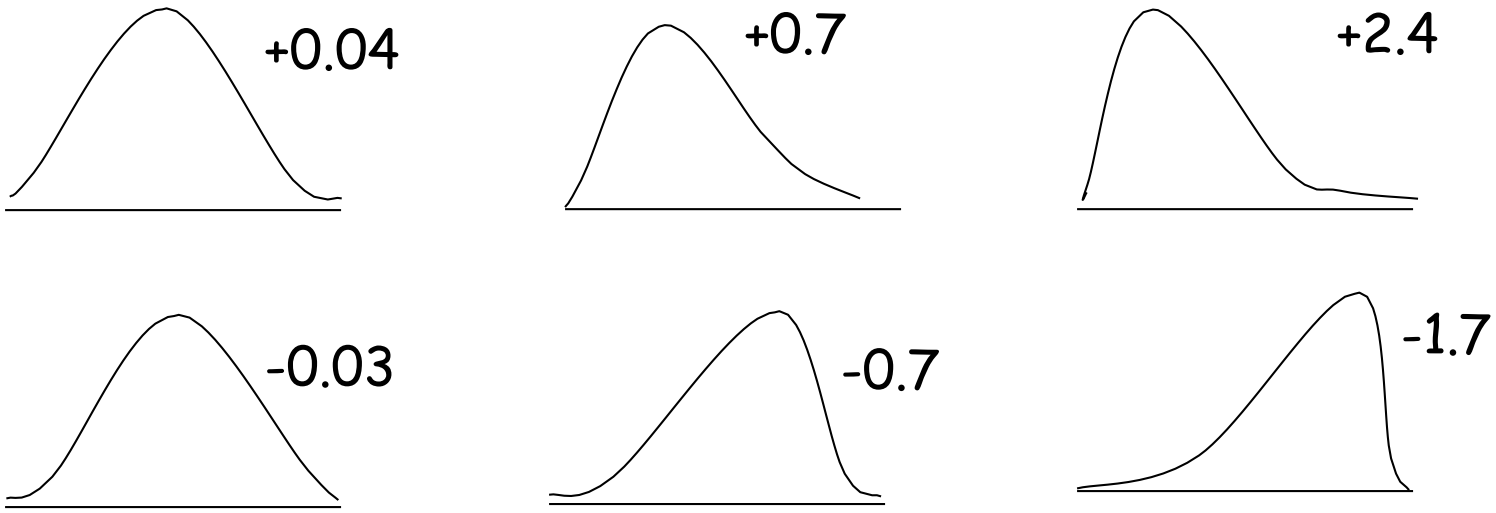
- If the measurement units are different then the standard deviations of two variables cannot be compared
- The standard deviation and mean of a variable are expressed in the same units, so taking the ratio of these two allows the units to be canceled.
- This ratio can then be compared to other such ratios in a meaningful way between two variables.
- The variable with the smaller CV is less dispersed than the variable with the larger CV. When the mean of a variable is zero the CV cannot be calculated.
- Even if the mean of a variable is not zero, but the variable contains both positive and negative values and the mean is close to zero, then the CV can be misleading
- The CV of a variable can be considered as a reasonable measure if the variable contains only positive (or negative) values

Skewness

- One measure of skewness is the Pearson's coefficient of skewness

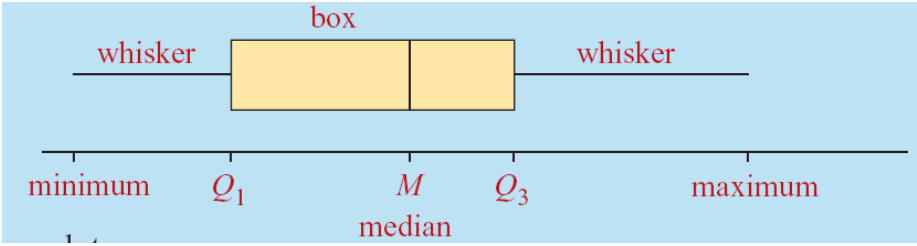
$$Pearson's\ Coefficient\ of\ Skewness = \frac{(Mean - Mode)}{Standard\ Deviation} = \frac{3(Mean - Median)}{Standard\ Deviation}$$

- Generally, skewness can take any value between +3 and -3



Box-and- Whiskers Plot

- Five Number Summary
  - List of the **minimum, Q_1, median, Q_3 and maximum**
- Graphical version of a five number summary
- Very useful for comparing distributions
- Can drawn either vertically or horizontally
- Any point **more than 1.5xIQR away** from the end of the box are classified as possible **outliers**.



- When we interpreting a box plot have to explain;

  Shape (symmetric or not),

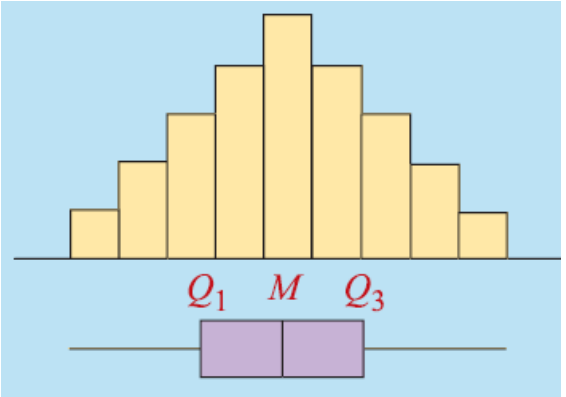
  Outliers (if any),

  Center using the median,

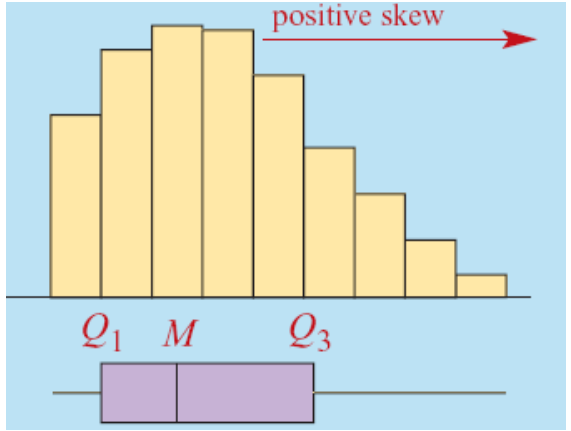
  Spread using the IQR or the Range

Relating a Box-Plot to Distribution Shape

- Symmetric distribution:
  - box plot is also symmetric.
  - Median in the middle of the box
  - Whiskers are equal size

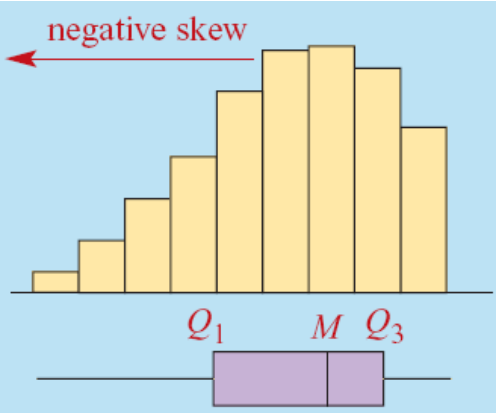


- Positively skewed distribution:
  - Median off-centre and generally to the left.
  - Left-hand whisker short,
  - Right-hand whisker long
  - reflecting the gradual tailing off of data values to the right.

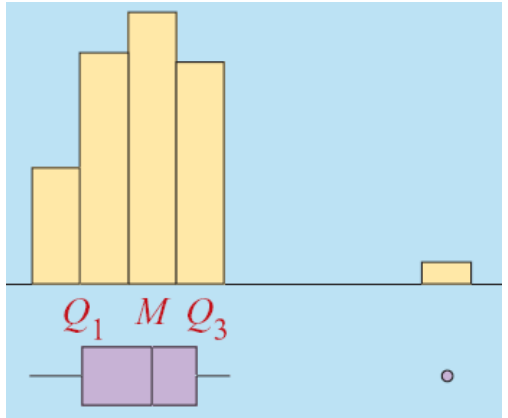


Relating a Box-Plot to Distribution Shape…

- Negatively skewed distribution:
  - Median off-centre and generally to the right.
  - Right-hand whisker short,
  - Left-hand whisker long
  - reflecting the gradual tailing off of data values to the left.



- Distributions with outliers
  - gaps between the main body and data in tails.



Measures Of Relationships

Covariance

- Extend to which two random variables vary together (co-vary)
- For two continuous random variables X and Y;

$$Cov(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

- Zero covariance: does not implies that variables are independent, there may be a nonlinear relationship

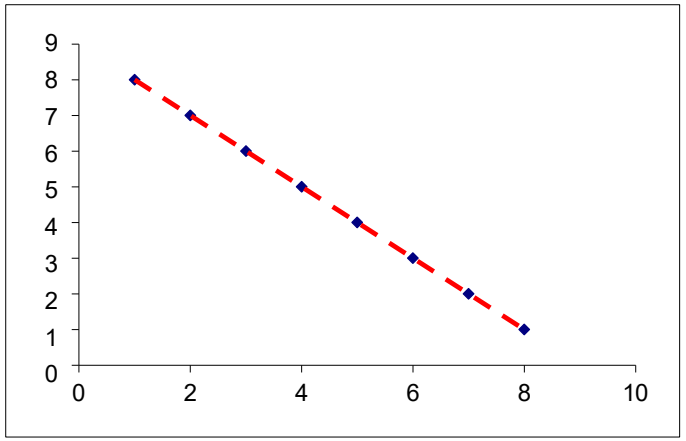
Correlation

- The strength of a linear relationship
- It indicates how closely the points in the scatter plot fit a straight line
- Coefficient of correlation, *r*, developed by Carl Pearson

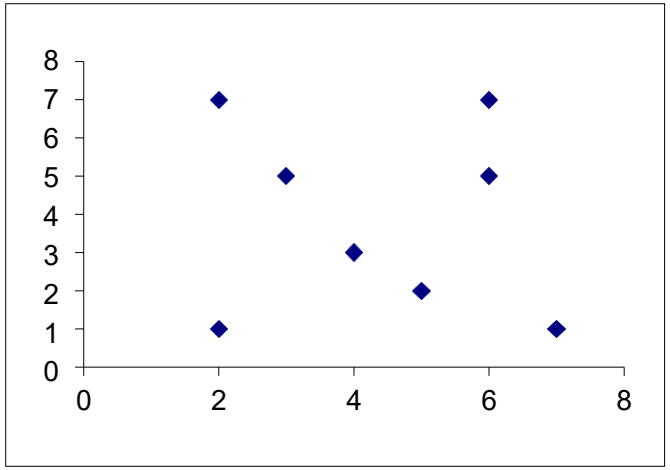
$$-1 \le r \le 1 \quad \text{or} \quad |r| \le 1$$

- If there is no fit, we say there is no relationship; points are scattered randomly on the plot, $r = 0$.
- If points lie exactly on a straight line, we say that there is a perfect linear relationship, $r = 1$ or $r = -1$.

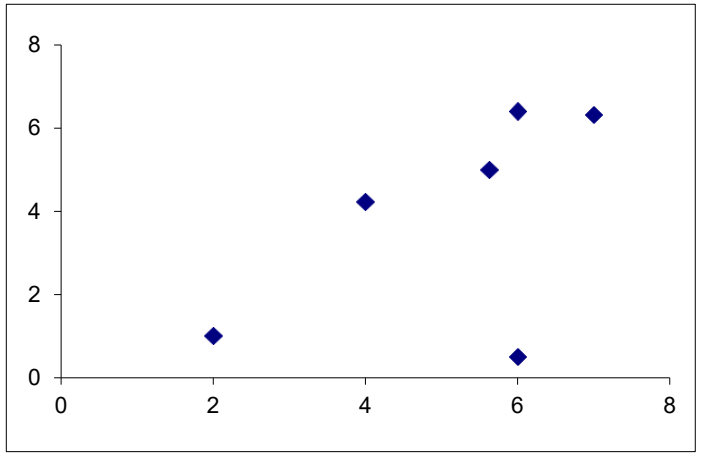
Correlation



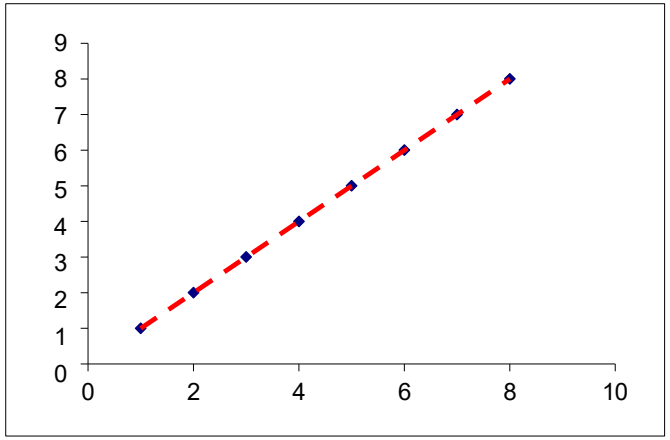
Perfect negative, $r = -1$



None, $r = 0$ or r is very close to 0



Positive strong, $r = 0.96$



Perfect positive, $r = 1$

Correlation

Guidelines for classifying the strength of a linear relationship

- Perfect positive, $r = +1$
- Strong positive, r between 0.75 and 0.99
- Moderate positive, r between 0.5 and 0.74
- Weak positive, r between 0.25 and 0.49
- No relationship, r between -0.24 and +0.24
- Weak negative, r between -0.25 and -0.49
- Moderate negative, r between -0.5 and -0.74
- Strong negative, r between -0.75 and -0.99
- Perfect negative, $r = -1$

Correlation

- We assume that:
    - The variables are numeric
    - The relationship is linear
    - There are no outliers in the data
- Can be misleading if there are outliers
- Covariance is a measure of co-variability between two variables.
- Correlation is just the translation of covariance into a unit-less measure that can understand (-1.0 to 1.0)

Calculating Pearson's Correlation Coefficient

- r gives a numerical measure of the degree of points in the scatter plot tend to cluster around a straight line.

$$r = \frac{Cov(X,Y)}{\sqrt{V(X)V(Y)}} = \frac{\sum (x-\bar{x})(y-\bar{y})}{S_x S_y}$$

where

$$s_x = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}} \qquad s_y = \sqrt{\frac{\sum (y-\bar{y})^2}{n-1}} \qquad \bar{x} = \frac{\sum x}{n} \qquad \bar{y} = \frac{\sum y}{n}$$

Coefficient of Determination

- The degree to which one variable can be predicted from another linearly related variables
- It can be calculated by squaring the correlation coefficient

$$\textbf{coefficient of determination} = r^2$$

- Percentage variation in the dependent variable that is explained by the variation in the independent variable.
- Practical interpretation of coefficient of determination
  - About $100(r^2)\%$ of the sample variance in dependent variable can be explained using x to predict in the straight-line model

Any questions?

Answers?

1. Calculate mean and standard deviation for the sample data, 10 12 14 15 17 18 18 24.
2. What are five number summary?
3. What measure from central tendency measures is suitable when the sample data consist of extreme values?
4. When the distribution of data is symmetric, what can you say about the mean and median of the data?
5. Fitness coach wants to understand how weekly training hours (X) relate to her clients' endurance test scores (Y). He collects data from 8 clients:

   Weekly Training Hours (X): 1, 2, 3, 4, 5, 6, 7, 8

   Endurance Test Scores (Y): 3, 5, 6, 9, 10, 13, 14, 17.

   Calculate the Pearson correlation coefficient between weekly training hours and endurance test score.